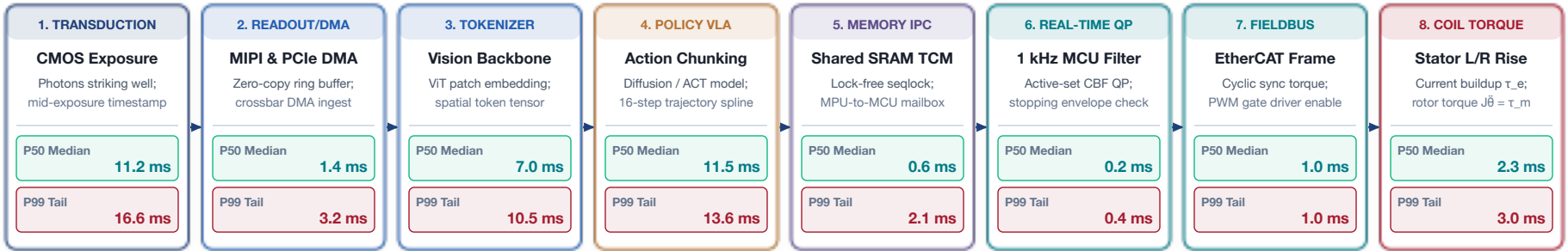


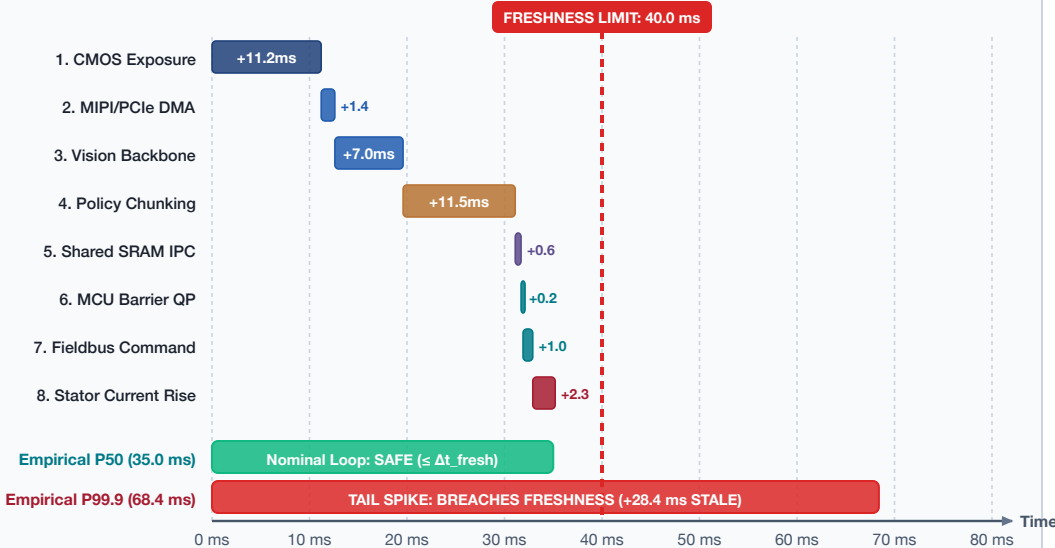
END-TO-END SENSE-TO-ACTUATION LATENCY WATERFALL & FRESHNESS BUDGET

Cumulative Wall-Clock Decomposition: From Photon Transduction to Torque Generation vs. the Freshness Deadline



CUMULATIVE SENSE-TO-ACTUATION DELAY WATERFALL (Δt)

Direct Empirical Sum vs. Measured Tail Convolution on Physical Hardware



PHYSICAL IMPACT: CLEARANCE COLLAPSE

Blind Coasting Travel: $d_{\text{react}} = v_0 \cdot \Delta t$

Operating velocity $v_0 = 1.2$ m/s; clearance $D_{\text{clear}} = 20.0$ cm

CASE A: Nominal P50 ($\Delta t = 35.0$ ms)

d_react 4.2cm Braking 7.2cm Margin +8.6cm ✓

Total d_stop = 11.4 cm $\leq$ 20.0 cm (Certified Safe Operation)

CASE B: Tail Latency P99.9 ($\Delta t = 68.4$ ms)

d_react 8.2cm (+4.0cm!) Braking 7.2cm Margin 4.6cm ⚠

Total d_stop = 15.4 cm (Safety buffer eroded by 46.5%)

FIRST-PRINCIPLES SYSTEMS IMPLICATION

- Latency is not an elastic software metric; it directly maps into unguided physical travel.
- P99.9 tail latency dictates true safe speed limits, not average computational throughput (P50).
- Stale observations ($\Delta t > \Delta t_{\text{fresh}}$) inject phase lag that destabilizes high-gain feedback loops.